

The Coin's Price

The Demon's Ledger Closes at In 2

Szilard-Landauer Closure of the Feedback Ledger

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Ch.74 gave the demon one fair bit and measured the ledger it moves. Here the bank closes the account: the same bit is a coin of measured face value ($I = \ln 2$, Szilard 1929) that costs $\ln 2$ to erase (Landauer 1961; measured in a colloidal particle by Berut et al., Nature 483, 187, 2012). We run the identical round-trip instrument ($\lambda 1 \rightarrow 2 \rightarrow 1$, $D = \beta = 1$, $\Delta F = 0$) with the erasure bill in the books. Measured: net cost $W_{\text{net}} = \langle W \rangle + \ln 2 > 0$ on every row of the frontier, including the dead coin (sign bit: $J = 0.99985$, $W_{\text{net}} = +0.804$) and the engaged bit (J rises to 1.1368, harvest to +0.0390, yet W_{net} closes from +0.736 to +0.654, never below zero). The ideal Szilard corner - extraction $\rightarrow \ln 2$, erasure $\rightarrow \ln 2$ - is the 0/0 of the information engine, reached only as the power fades, exactly as in Carnot's corner of Ch.73. Removable value: 0. Books close with the coin.

1. The Coin and the Bill

One fair bit carries $I = \ln 2$ nats of information (Szilard 1929): a recorded measurement that, fed causally to the return protocol, lifts the exponential ledger toward e^I (Ch.74, Sagawa-Ueda 2010). No information is free to keep: the same two-state memory costs $k_B T \ln 2$ to erase (Landauer 1961). Berut et al. measured the bill in a colloidal particle caught in a double-well trap - $0.69 k_B T$ per bit at slow rates, exactly $\ln 2$. Toyabe et al. (2010) measured the coin spending its face value in a feedback bead rotor: roughly $\ln 2 k_B T$ of work per bit. The demon therefore cannot win on any complete cycle that returns its memory to a standard state.

$$\text{coin face value } I = \ln 2 = 0.6931 \quad ; \quad \text{erasure bill } E = \ln 2 = 0.6931$$

2. The Deuteronomy of the Coin (Second Law)

Two measured ceilings hold on the same instrument, and the third when the memory is closed:

- $J = \langle e^{(-W)} \rangle \leq e^I = 2.000$ (Sagawa-Ueda 2008/2010) - the ledger moved by causal information.
- $W_{\text{net}} = \langle W \rangle + E \geq 0$ (Bennett 1982) - extraction must beat the erasure bill; measured below on every row.
- No cycle extracts; the residual W_{net} is the price of speed, vanishing only at zero power.

The two are one statement twice: the exponential ledger never crosses e^I , and the mean book never closes with a profit once the coin is returned. Information is the interest, not the principal.

3. The Dead Coin: a Bank Note, Unread

The sign bit is a real, fair coin carrying $I = \ln 2$, spent in full on the return speed. The trap is reflection-symmetric, so the sign predicts nothing (Ch.74):

$$\begin{aligned} [2] \text{ sign bit } \rightarrow \text{ speed: } J &= 0.99985, \ln J = -0.00015, \langle W \rangle = +0.11069 \\ W_{\text{net}} &= \langle W \rangle + \ln 2 = +0.80383 \quad (\text{the dead coin: } 0.693 \text{ spent for nothing}) \end{aligned}$$

A bank note, unread, still counts against the account. The null meter of Ch.74 - an uncorrelated bit leaving $J = 1$ - now carries a price: the control run with no coin costs only $\langle W \rangle = +0.113$. The demon that pays for silence spends 0.69 and harvests 0.

4. The Frontier with the Bank

Each fair bit buys a return protocol (fast for far, slow for near). Measured across the leverage frontier (Heun SRK2, seed 42, 1.5M+ runs):

fast/slow	J	$\ln J$	$\langle W \rangle$	$H = -\langle W \rangle$	$W_{\text{net}} = \langle W \rangle + \ln 2$	$H / \ln 2$
0.35/2.00	1.07484	+0.07217	+0.04261	-0.04261	+0.73576	-0.061
0.25/4.00	1.09804	+0.09352	+0.01500	-0.01500	+0.70815	-0.022
0.15/6.00	1.11565	+0.10944	-0.00951	+0.00951	+0.68364	+0.014
0.10/8.00	1.12420	+0.11707	-0.02160	+0.02160	+0.67154	+0.031
0.05/16.0	1.13681	+0.12822	-0.03898	+0.03898	+0.65417	+0.056

The engaged bit lifts J toward e^I ($\ln J$ rises 0.072 $\rightarrow$ 0.128, never past 0.693) and the mean work turns negative past the crossover near fast 0.15/slow 6.0: the demon harvests. Yet W_{net} closes monotonically from +0.736 to +0.654 and never touches zero - the same noise lends the harvest and the bank recalls it with interest. Every payoff is a subsidy from the coin, denominated in $\ln 2$.

5. The 0/0: the Reversible Corner of the Information Engine

In the ideal limit - a quasistatic return perfectly matched to the measured bit - the extraction per bit approaches $\ln 2$ (Szilard's engine) while the erasure bill remains $\ln 2$ (Landauer's):

$$W_{\text{net}} \rightarrow 0 \text{ as } (\text{extraction } \rightarrow \ln 2, \text{ erasure } \rightarrow \ln 2): \quad 0/0, \text{ removable value } 0$$

Measured W_{net} declines along the frontier (0.736 $\rightarrow$ 0.654) and $H / \ln 2$ rises toward 1 as the power fades - the identical shape of Ch.73's Carnot corner: efficiency approaches its ideal ceiling only at zero power. The information engine and the thermal engine are one portrait: the reversible corner is the removable singularity, and the coin that would price it is the bill that closes it.

Main Results

THEOREM

The Coin Closes the Ledger: Let the round trip ($\Delta F = 0$) be run with a fair bit ($I = \ln 2$) fed causally to the return speed. Then $J =$

$\langle e^{-W} \rangle \leq e^{-I}$ (Sagawa-Ueda) and $W_{\text{net}} = \langle W \rangle + \ln 2 \geq 0$ (Bennett-Landauer). Measured: J up to 1.1368 ($\ln J$ 0.128), W_{net} from +0.736 to +0.654, never negative; dead coin $W_{\text{net}} = +0.804$.

COROLLARY

The 0/0 Resolution: the ideal Szilard corner (extraction $\rightarrow \ln 2$, erasure $\rightarrow \ln 2$) gives $W_{\text{net}} = 0/0$ with removable value 0 - a payoff exists only as the power vanishes, exactly Carnot's corner (Ch.73).

COROLLARY

The Dead Coin Is Priced: information uncorrelated with the work leaves $J = 1$ but still bills $\ln 2$ - measured $W_{\text{net}} = +0.804$ vs control +0.113.

COROLLARY

Work from Information Is Work from the Coin: at best the harvest is the repayment of the erasure bill, capped per bit (Toyabe et al. 2010).

Conclusion

Ch.74 showed the demon moves the ledger; this chapter shows the bank moves it back. Every bit the demon spends is minted at $\ln 2$ and recalled at $\ln 2$; the measured frontier is a lifelong overdraft - J creeps toward e^{-I} , the harvest toward the coin, and the books toward zero, which they touch only at the quiet corner where no power flows. The removable singularity of the information engine is removed by its own currency: the coin's price is the bill's value, and the demon settles at par.

References

- [1] Szilard, L. (1929). Z. Phys. 53, 840.
- [2] Landauer, R. (1961). IBM J. Res. Dev. 5, 183.
- [3] Bennett, C.H. (1982). Int. J. Theor. Phys. 21, 905.
- [4] Berut, A., Arakelyan, A., Petrosyan, A., Ciliberto, S., Dillenschneider, R. & Lutz, E. (2012). Nature 483, 187.
- [5] Toyabe, S., Sagawa, T., Ueda, M., Munakata, E. & Sano, M. (2010). Nature Phys. 6, 988.
- [6] Sagawa, T. & Ueda, M. (2008). Phys. Rev. Lett. 100, 080603.
- [7] Sagawa, T. & Ueda, M. (2010). Phys. Rev. Lett. 104, 090602.
- [8] Parrondo, J.M.R., Horowitz, J.M. & Sagawa, T. (2015). Nature Phys. 11, 131.
- [9] Jarzynski, C. (1997). Phys. Rev. Lett. 78, 2690.
- [10] Seifert, U. (2012). Rep. Prog. Phys. 75, 126001.
- [11] Esposito, M. & Van den Broeck, C. (2011). EPL 95, 40004.